

1 Materials and Methods

1.1 The Long-Term Phosphorus Fertilization Experiment

The soil samples for this thesis originate from a set of six long-term field trials in Switzerland, established by Agroscope between 1989 and 1992. The primary objective of these experiments was to validate and re-evaluate Swiss phosphorus (P) fertilization guidelines by assessing long-term crop yield responses to varying P inputs across different pedoclimatic conditions. A detailed description of the experimental design and site characteristics can be found in Hirte et al. (2021).

The experiment was set up as a **completely randomized block design** with four field replications at each site. The core of the experiment consists of six fixed-plot treatments representing different P fertilization levels, which were applied annually as superphosphate before tillage and sowing. These levels were based on percentages of the officially recommended P inputs: 0% (Zero), 33% (Deficit), 67% (Reduced), 100% (Norm), 133% (Elevated), and 167% (Surplus).

1.2 Experimental Sites

The six experimental sites are located in the main crop-growing regions of Switzerland: **Rümlang-Altwi (ALT)**, **Cadenazzo (CAD)**, **Ellighausen (ELL)**, **Grabs (GRA)**, **Oensingen (OEN)**, and **Zurich-Reckenholz (REC)**. The key soil properties are summarized below.

```
sites_df_corrected <- data.frame(  
  Site = c("ALT", "CAD", "ELL", "GRA", "OEN", "REC"),  
  `Soil Type (WRB)` = c("Calcaric Cambisol", "Eutric Fluvisol", "Eutric Cambisol", "Calcaric Fluvisol",  
    "Eutric Gleysol", "Eutric Gleysol"),  
  `Clay (%)` = c(22, 8, 33, 17, 37, 39),  
  `Sand (%)` = c(48, 40, 31, 34, 32, 25),  
  `Organic C (g/kg)` = c(21, 14, 23, 16, 24, 27),  
  `pH (H2O)` = c(7.9, 6.3, 6.6, 8.3, 7.1, 7.4),  
  check.names = FALSE  
)  
  
knitr::kable(sites_df_corrected)
```

Table 1: Soil characteristics of the six long-term experimental sites. Data adapted from Hirte et al. (2021).

Site	Soil Type (WRB)	Clay (%)	Sand (%)	Organic C (g/kg)	pH (H2O)
ALT	Calcaric Cambisol	22	48	21	7.9
CAD	Eutric Fluvisol	8	40	14	6.3
ELL	Eutric Cambisol	33	31	23	6.6
GRA	Calcaric Fluvisol	17	34	16	8.3
OEN	Gleyic-calc. Cambisol	37	32	24	7.1
REC	Eutric Gleysol	39	25	27	7.4

Soil samples for this thesis were collected in the year 2022 from the topsoil layer (0-20 cm).

1.3 Phosphorus Desorption Kinetics and Thermodynamics

The laboratory extraction of phosphorus was based on the sequential kinetic principles established by Flossmann & Richter (1982), subsequently modified to capture high-resolution temporal kinetics and corrected to reflect true thermodynamic activity.

1.3.1 Adapted Kinetic Protocol for This Study

To capture the desorption process with high temporal resolution while preserving the native soil matrix, a modified water-extraction protocol was employed:

1. **Pre-washing to Remove Soluble P:** 10 g of air-dried soil was suspended in 200 ml of deionized water and shaken for 60 minutes at 120 Hz. The suspension was centrifuged for 15 minutes at 4000 rpm, and the supernatant containing the readily soluble P was discarded.

2. **Kinetic Extraction:** The remaining soil pellet was resuspended in 200 ml of fresh deionized water. The suspension was shaken continuously, and subsamples were taken at eight time points: 2, 4, 10, 15, 20, 30, 45, and 60 minutes.
3. **Analysis:** Subsamples were immediately filtered, and the stoichiometric concentration of orthophosphate (c_i) was determined colorimetrically using the malachite green method (Van Veldhoven & Mannaerts, 1987).

1.3.2 Thermodynamic Correction to Chemical Activity

Although the kinetic extraction utilized deionized water, the immediate dissolution of native soil salts (Ca^{2+} , Mg^{2+} , K^+ , etc.) generates a non-zero ionic strength (I) in the suspension. Because plant roots respond to thermodynamic energy rather than physical mass, the stoichiometric concentration of the extract (c_i) must be corrected to its true chemical activity (a_i).

The activity coefficient (γ) for the divalent phosphate ion (HPO_4^{2-} , $z = -2$) was calculated for each site-specific matrix using the **Davies Equation**, which is highly robust for agricultural soil solutions up to $I \approx 0.1$ M:

$$\log_{10}(\gamma) = -0.5 \cdot z^2 \left(\frac{\sqrt{I}}{1 + \sqrt{I}} - 0.3I \right)$$

The true thermodynamic activity (a_i) of the phosphorus in the extract was then calculated as the product of the raw stoichiometric concentration and the activity coefficient:

$$a_i = c_i \cdot \gamma$$

This correction neutralizes pedochemical variances caused by differing native background salinities and pH levels across the STYCS sites, converting the raw data into a universal thermodynamic Intensity (I) metric.

1.4 Statistical Analysis and Modeling

All data processing, mathematical modeling, and visualization were conducted using the R programming language (v. 4.3.1) (R Core Team, 2022). Primary packages included **nlme** (Pinheiro et al., 2022) for non-linear kinetic modeling, **lme4** (Bates et al., 2015) for linear mixed-effects modeling, and **deSolve** and **ReacTran** for partial differential equation solving.

1.4.1 1. Modeling of Desorption Kinetics and the Activity Flux (J_0)

To derive the kinetic parameters, a non-linear mixed-effects model (**nlme**) was fitted to the exact solution of the first-order rate equation, utilizing the thermodynamically corrected activities:

$$a(t) = a_{max} \times (1 - e^{-k \times t'})$$

Where $a(t)$ is the P activity at time t , a_{max} is the maximum thermodynamic activity of the desorbable P pool (the asymptote), k is the first-order desorption rate constant, and t' is an adjusted time ($t_{min} + 3$ min) accounting for initial dissolution. Sample-specific deviations were modeled as random effects to capture the unique desorption characteristics of each soil.

From these parameters, a novel predictor for agronomic availability, the **Initial Activity Flux (J_0)**, was defined mathematically as the product of the rate constant and the maximum thermodynamic activity:

$$J_0 = k \cdot a_{max}$$

1.4.2 2. Thermodynamic $Q(I)$ Isotherms and Buffering Capacity

To deduce the Phosphorus Buffering Capacity (PBC), a thermodynamic Quantity-Intensity (Q/I) framework was constructed. The P_{AAE10} extraction was utilized as the standard mass Quantity (Q , mg P kg⁻¹), while the Davies-corrected a_{max} was utilized as the native Intensity (I , μ M).

The relationship between these variables across the varying fertilizer treatments at each site was parameterized using a log-linearized **Freundlich Isotherm**:

$$Q = K_f \cdot I^{\frac{1}{n}}$$

The true thermodynamic PBC for each specific site and state was then calculated analytically as the first derivative of the fitted Freundlich isotherm with respect to Intensity:

$$\text{PBC} = \frac{dQ}{dI} = K_f \cdot \frac{1}{n} \cdot I^{(\frac{1}{n}-1)}$$

1.4.3 3. Agronomic Normalization: The Historical Baseline Strategy

To statistically evaluate the predictive power of the chemical parameters across six highly diverse pedoclimatic sites, raw crop yield and P-uptake data required robust normalization to isolate chemical P-limitation from overarching climatic noise.

Standard normalization (dividing by the maximum yield of a given site within the specific experimental year) was rejected, as it artificially masks broad climatic limitations (e.g., drought) and forces the best plot of a poor year to equal 100%, mathematically erasing the biological penalty. Instead, a **historical biological baseline strategy** was employed.

Using the 40-year STYCS historical dataset, the true biological potential (Y_{max} and $Uptake_{max}$) was calculated for each specific site and crop species by taking the 95th percentile of all fully fertilized (100% and 167% P) treatments. Current experimental yields and uptake values were divided by this historical baseline to calculate **Relative Historical Yield** (Y_{rel_hist}), providing a highly stable metric representing the percentage of the crop's long-term biological potential achieved.

1.4.4 4. Comparative Modeling of Agronomic Outcomes

To test the core hypothesis, two competing sets of linear mixed-effects models (**lmer**) were constructed to predict Y_{rel_hist} and Relative P-Uptake. Both sets shared an identical random effects structure—(1|year) + (1|Site) + (1|Site:block)—to absorb unmeasured pedoclimatic and meteorological variance.

- **Model 1: The Kinetic/Thermodynamic Approach:** Utilized the novel Activity Flux (J_0) as the primary fixed effect, testing the hypothesis that dynamic supply rate dictates plant availability.
- **Model 2: The Standard STP (GRUD) Approach:** Utilized the standard Swiss static soil tests (P_{CO2} and P_{AAE10}) and their interaction as fixed effects.

```
variables_df <- data.frame(
  Abbreviation = c("$Y_{rel}\\_hist}$", "$Uptake_{rel}$", "$P_{bal}$",
    "$k$", "$a_{max}$", "$J_0$",
    "$P_{AAE10}$", "$\\text{PBC}$"),
  `Variable` = c("Relative Yield (Historic)", "Relative P Uptake", "Cumulative P Balance",
    "Rate Constant", "Maximum Activity (Intensity)", "Activity Flux",
    "Standard Mass (Quantity)", "Phosphorus Buffering Capacity"),
  Unit = c("unitless", "unitless", "kg P ha-1",
    "min-1", "μM", "μM min-1",
    "mg P kg-1", "mg kg-1 μM-1"),
  Description = c(
    "Plot yield normalized by the site/crop-specific 95th percentile of historical well-fertilized y",
    "Plot P-uptake normalized by historical maximum uptake for the specific site and crop.",
    "Long-term net P budget (inputs minus outputs).",
    "First-order rate constant of P desorption.",
    "Davies-corrected thermodynamic activity asymptote of the water-desorbable P pool.",
```

```

    "Initial thermodynamic supply rate, calculated as $k \\times a_{max}$.",
    "Physical mass of extractable P (AAE10 method), acting as the Q-factor.",
    "The mathematical derivative ($dQ/dI$) of the site-specific Freundlich isotherm."
  )
)
knitr::kable(variables_df)

```

Table 2: Description of key variables used in the thermodynamic and agronomic models.

Abbrevia- tion	Variable	Unit	Description
Y_{rel_hist}	Relative Yield (Historic)	unitless	Plot yield normalized by the site/crop-specific 95th percentile of historical well-fertilized yields.
$Uptake_{rel}$	Relative P Uptake	unitless	Plot P-uptake normalized by historical maximum uptake for the specific site and crop.
P_{bal}	Cumulative P Balance	kg P ha ⁻¹	Long-term net P budget (inputs minus outputs).
k	Rate Constant	min ⁻¹	First-order rate constant of P desorption.
a_{max}	Maximum Activity (Intensity)	μM	Davies-corrected thermodynamic activity asymptote of the water-desorbable P pool.
J_0	Activity Flux	μM min ⁻¹	Initial thermodynamic supply rate, calculated as $k \times a_{max}$.
P_{AAE10}	Standard Mass (Quantity)	mg P kg ⁻¹	Physical mass of extractable P (AAE10 method), acting as the Q-factor.
PBC	Phosphorus Buffering Capacity	mg kg ⁻¹ μM ⁻¹	The mathematical derivative (dQ/dI) of the site-specific Freundlich isotherm.

1.4.5 5. Mechanistic Modeling of the Rhizosphere (Barber-Cushman Modification)

To mechanistically validate the field-scale statistical findings and visualize the limitations of static buffering capacity, a modified radial transport model based on Barber-Cushman principles was constructed. Standard iterations of the Barber-Cushman model calculate radial diffusion using stoichiometric concentration (C_l) and a static buffer power ($b = \Delta Q / \Delta I$), effectively assuming instantaneous soil-solution equilibrium.

To reflect the kinetic bottlenecks identified in this study, the partial differential equation (PDE) for radial transport in cylindrical coordinates was modified. The driving gradient was shifted to thermodynamic activity (a_l), and the static buffer power was replaced with a dynamic, first-order kinetic source term ($R_{desorption}$):

$$\frac{\partial a_l}{\partial t} = D_e \left(\frac{\partial^2 a_l}{\partial r^2} + \frac{1}{r} \frac{\partial a_l}{\partial r} \right) + \frac{r_0 v_0}{r} \frac{\partial a_l}{\partial r} + R_{desorption}$$

Where D_e is the effective diffusion coefficient, r is the radial distance, r_0 is the root radius, and v_0 is the water flux at the root surface. The kinetic source term resupplying the solution was defined using the experimentally derived parameters:

$$R_{desorption} = k \cdot (a_{max} - a_l)$$

Furthermore, the inner boundary condition at the root surface ($r = r_0$) was upgraded to reflect a dual-affinity Michaelis-Menten uptake system, accounting for both high-affinity (PHT1) and low-affinity (LATS) transporter kinetics responding directly to chemical activity:

$$J_r = \frac{V_{max1} \cdot (a_l - a_{min})}{K_{m1} + (a_l - a_{min})} + \frac{V_{max2} \cdot (a_l - a_{min})}{K_{m2} + (a_l - a_{min})}$$

The modified PDEs were solved numerically in R utilizing the **ReacTran** and **deSolve** packages to generate dynamic spatial profiles of the P-depletion zone over time.

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